Exercise 5.1.1 Show that Eq. (5.1.9) may be written as an integral over E and a sum over the ± index as

$$U(t) = \sum_{a=\pm} \int \left[\frac{m}{\sqrt{2mE}}\right] |E,\alpha\rangle \langle E,\alpha| \, e^{-iEt/h} dE$$

Equation 5.1.9 is the propagator for a free particle written as

$$U(t) = \int_{-\infty}^{\infty} |p\rangle \langle p| \, e^{-ip^2 t/2m\hbar} dp \tag{5.1.9}$$

From the time independent Schrödinger equation $H|E\rangle = E|E\rangle = \frac{P^2}{2m}|E\rangle$ (5.1.3), we find that allowed values of p of a free particle are $p = \pm\sqrt{2mE}$. Thus for every eigenvalue E, there corresponds a two-fold degenerate eigenspace defined by p_+ and p_-. The propagator is defined as an integral over all stationary states, accounting for any degeneracies from compatible observables, with an attached E dependent phase term (4.3.13).

We can substitute $|p\rangle = |E,\alpha\rangle$ provided we use the change of variable $p^2 = 2mE$, and $dp = (2mE)^{-1/2}dE$. From the change of variables, $p \in [-\infty,\infty]$ corresponds to $E \in [0,\infty]$ and $\alpha = \pm$, thus we arrive at the sum and integral of the stated expression.

Exercise 5.1.2 By solving the eigenvalue equation (5.1.3) in the X basis, regain equation (5.1.8), i.e. show that the general solution of energy E is

$$\psi_E(x) = \beta \frac{\exp\left(i\sqrt{2mE}x/h\right)}{\sqrt{2\pi\hbar}} + \gamma \frac{\exp\left(-i\sqrt{2mE}x/h\right)}{\sqrt{2\pi\hbar}}$$

Is this function in Hilbert space if $E < 0$?

Projecting $\langle x|$ onto (5.1.3), we get

$$\langle x| H |E\rangle = \langle x| E |E\rangle = \langle x| \frac{P^2}{2m} |E\rangle$$

on the LHS

$$E\langle x|E\rangle = E\psi_E(x)$$

on the RHS, and representing P^2 in the x-basis, and inserting the identity of $I = \int dx' |x'\rangle \langle x'|$ inbetween P^2 and $|E\rangle$, we arrive at

$$\frac{P^2}{2m}|E\rangle = \frac{-\hbar^2}{2m}\frac{d^2}{dx^2}\psi_E(x)$$

The resulting differential equation $\psi_E''(x) = -\frac{2mE}{\hbar^2}\psi_E(x)$ is generally solved by $\psi_E(x) = \beta\exp(ikx) + \gamma\exp(-ikx)$ where $k = \sqrt{2mE}/\hbar$. Where β and γ are fixed by the initial conditions of the problem. We can extract the common denominator $\sqrt{2\pi\hbar}$ from both factors in order to regain the desired expression. Lastly, the function is not contained in Hilbert space if $E < 0$. Under that condition, the exponentials will have a real argument, and as $x \to \pm\infty$, $\psi_E(x)$ blows up to infinity. This violates the condition that Hilbert space requires square integrable functions, i.e. the function can be normalized when integrated over all space.

Exercise 5.1.3 Consider a problem involving the time evolution of a gaussian packet. Consider the initial state with initial momentum $p_0 = 0$, starting time $t' = 0$,and the width $\Gamma = 1$

$$\psi(x,0) = \frac{1}{(\pi)^{1/4}} \exp(-x^2/2)$$

. Find $\psi(x,t)$ (5.1.15) using $U(t) = e^{-iHt/\hbar}$ instead of finding $U(t)$ in the x-basis, which is what is found in the textbook. Let $\alpha = (i\hbar t/2m)$, Then this construction of U can be expressed as

$$U(t) = \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{i\hbar t}{2m}\right)^n \left(\frac{d}{dx}\right)^{2n} = \left(1 + \alpha \frac{d^2}{dx^2} + \frac{1}{2!}\alpha^2 \frac{d^4}{dx^4} + \frac{1}{3!}\alpha^3 \frac{d^6}{dx^6}\right)$$

First we can express $\psi(x,0)$ as a power series

$$\psi(x,0) = \pi^{-1/4} \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{n! 2^n} = \pi^{-1/4}\left(1 - \frac{x^2}{1!2} + \frac{x^4}{2!4} - \frac{x^6}{3!8} + \frac{x^8}{4!16} - \frac{x^{10}}{5!32} + ...\right)$$

$\psi(x,t)$ is equal to $U(t)$ on $\psi(x,0)$ (If I did this in some basis, then I would have to integrate). To illustrate a few terms, consider each $U_n(t)$ where $U(t) = \sum_{n=0}^{\infty} U_n(t)$, and the action of $U_n(t)\psi(x,0)$.

For n=0:

$$U_0(t)\psi(x,0) = \psi(x,0)$$

For n=1:

$$U_1(t)\psi(x,0) = \alpha\pi^{-1/4}\left(-\frac{2}{1!2} + \frac{(4\cdot 3)x^2}{2!4} - \frac{(6\cdot 5)x^4}{3!8} + \frac{(8\cdot 7)x^6}{4!16} - ...\right)$$

For n=2:

$$U_2(t)\psi(x,0) = \alpha^2\pi^{-1/4}\frac{1}{2!}\left(\frac{(4\cdot 3\cdot 2\cdot 1)}{2!4} - \frac{(6\cdot 5\cdot 4\cdot 3)x^2}{3!8} + \frac{(8\cdot 7\cdot 6\cdot 5)x^4}{4!16} - ...\right)$$

for n=3:

$$U_3(t)\psi(x,0) = \alpha^3\pi^{-1/4}\frac{1}{3!}\left(-\frac{6!}{3!8} + \frac{8!x^2}{2!4!16} - \frac{10!x^4}{4!5!32} + ...\right)$$

If I collect terms, $U(t)\psi(x,0)$ can be re-expressed as,

$$\begin{aligned} U(t)\psi(x,0) = \pi^{-1/4}&\left(1 - \alpha + \frac{3}{2}\alpha^2 - \frac{5}{2}\alpha^3 + ...\right)x^0 \\ &\pi^{-1/4}\left(-\frac{1}{2} + \frac{3}{2}\alpha - \frac{15}{4}\alpha^2 + \frac{35}{4}\alpha^3 - ...\right)x^2 \\ &\pi^{-1/4}\left(\frac{1}{8} - \alpha\frac{5}{2} + \alpha^2\frac{35}{4} - \alpha^3\frac{105}{16} + ...\right)x^4 + ... \end{aligned}$$

Note that, if $\alpha = \beta/2$, we can use for $n \in [0,1,2,...]$

$$\left(1+\beta\right)^{-n-1/2} = 1 - (n+1/2)\beta + \frac{(n+1/2)(n+3/2)}{2!}\beta^2 + ...$$

Then $U(t)\psi(x,0)$ can be rewritten as

$$U(t)\psi(x,0) = \pi^{-1/4}\left[\left(1+\frac{i\hbar t}{m}\right)^{-1/2}x^0 - \frac{1}{1!2^1}\left(1+\frac{i\hbar t}{m}\right)^{-3/2}x^2 + \frac{1}{2!2^2}\left(1+\frac{i\hbar t}{m}\right)^{-5/2}x^4 - ...\right]$$

Finally, by factoring and contracting the remaining term using taylor expansion, then we arrive at the desired expression which compares to equation (5.1.15) from the text.

$$\psi(x,t) = U(t)\psi(x,0) = \pi^{-1/4}\left(1+\frac{iht}{m}\right)^{-1/2}\left[\sum_{n=0}^{\infty}(-1)^n\frac{x^{2n}}{n!2^n}\left(1+\frac{iht}{m}\right)^{-n}\right]$$

$$= \pi^{-1/4}\left(1+\frac{i\hbar t}{m}\right)^{-1/2}\exp\left(-\frac{x^2}{2(1+i\hbar t/m)}\right)$$

Exercise 5.1.4 Consider the wave function

$$\begin{cases}\psi(x,0) = \sin(\pi x/L) & ,|x| \le L/2\\ = 0 & ,|x| > L/2\end{cases}$$

It is clear that when thsi function is differentiated any number of times we get another function that is confined to $|x| < L/2$. Consequently the action of

$$U(t) = \exp\left[\frac{i}{h}\left(\frac{\hbar^2 t}{2m}\right)\frac{d^2}{dx^2}\right]$$

on this function is to give a function confined to $|x| < L/2$. What about the spreading of the wavepacket?

If we were to evaluate $\psi(x,t) = U(t)\psi(x,0)$ we would expand the operator power series to act on the original function and find that each derivative reproduces $\psi(x,0)$ with some additional phase factor in front. There exists a paradox here because we have a free particle with no potential barriers acting on it, and it will, for all time, remain contained between $|x| < L/2$. Typically, we would expect it to spread out beyond the boundaries of it's initial state (remember that even though the wave function has no explicit mention of momentum, a definite 0 momentum implies infinite spatial uncertainty. Since the spacial uncertainty is definite to $|x| < L/2$, there is an implied momentum distribution which can be found with a Fourier transform to the momentum conjugate basis).

But there is no paradox here because given an abrupt step jump in the wave function, the derivatives of ∂_x^2 always reproduce discontinuous functions are the boundaries (and hence repeated delta functions and higher derivatives of delta functions, this is called getting increasingly singular). The power series expansion of $U(t)$ is only defined as long as the series converges to a definite limit, which it does not when acting on this particular function.

Exercise 5.2.1 A particle is in the ground state of a box of length L. Suddenly the box expands symmetrically to twice its size, leaving the wavefunction undisturbed. Show that the probability of finding the particle in the ground state of the new box is $(8/3\pi)^2$

Following the particle in box (potential is nonzero around $|x| < L/2$. The independent schrödinger yields solutions of the following form,

$$\begin{cases}\psi_n(x) = \left(\frac{2}{L}\right)^{1/2}\sin\left(\frac{n\pi x}{L}\right) & ,n \text{ even}\\ \psi_n(x) = \left(\frac{2}{L}\right)^{1/2}\cos\left(\frac{n\pi x}{L}\right) & ,n \text{ odd}\end{cases} \tag{5.2.15}$$

The ground state for a particle in these boxes are as follows

$$\begin{cases} \psi_1(x) = \left(\frac{2}{L}\right)^{1/2} \cos\left(\frac{\pi x}{L}\right) & , \text{ Box width = L} \\ \phi_1(x) = \left(\frac{1}{L}\right)^{1/2} \cos\left(\frac{\pi x}{2L}\right) & , \text{ Box width = 2L} \end{cases}$$

The probability of finding $\phi_1(x)$ after an initial state of $\psi_1(x)$ is the following. Note that the limits are still reduced to only $|x| < L/2$ since $\psi_1(x)$ is zero otherwise.

$$P = |\langle\phi_1|\psi_1\rangle|^2 = \left|\int_{-L/2}^{L/2} \phi_1^*(x)\psi_1(x)dx\right|^2 = \left|\frac{\sqrt{2}}{L}\int_{-L/2}^{L/2} \cos(\pi x/L)\cos(\pi x/2L)dx\right|^2$$

Using the product of cosines,

$$P = \left|\frac{\sqrt{2}}{L}\int_{-L}^{L} \frac{1}{2}\Big(\cos\big(\frac{3\pi x}{2L}\big) + \cos\big(\frac{\pi x}{2L}\big)\Big)dx\right|^2 = \Big(\frac{8}{3\pi}\Big)^2$$

Exercise 5.2.2 a) Show that for any normalized $|\psi\rangle$, $\langle\psi| H |\psi\rangle \geq E_0$, where E_0 is the lowest energy eigenvalue.

Since E_0 is the lowest energy eigenvalue, let all eigenvalues be satisfied by the expression $E_n = E_0 + f(n)$, where $f(n)$ is positive. We define $a_n \equiv \langle E_n|\psi\rangle$ and then expand ψ in H's eigenbasis to see that,

$$\begin{aligned}\langle\psi| H |\psi\rangle &= \langle\psi| \sum_n H |E_n\rangle \langle E_n|\psi\rangle = \sum_{m,n} a_m^* a_n E_n \langle E_m|E_n\rangle = \sum_n |a_n|^2 E_n \\ &= \sum_n |a_n|^2 (E_0 + f(n)) = E_0 + \sum_n |a_n|^2 f(n) \geq E_0\end{aligned}$$

b) Prove the following theorem: Every attractive potential in one dimension has atleast one bound state. Hint: Since V is attractive, if we define $V(\infty) = 0$ it follows that $V(x) = -|V(x)|$ for all x. To show that there exists a bound state with $E < 0$, consider

$$\psi_a(x) = \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\alpha x^2/2}$$

and calculate

$$E(\alpha) = \langle\psi_a| H |\psi_a\rangle, \qquad H = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} - |V(x)|$$

Show that $E(\alpha)$ can be made negative by a suitable choice of α. The desired result follows from the application of the theorem proved above

Calculating $E(\alpha)$ can be divided into calculating $\langle T\rangle$ and $\langle V\rangle$. For the kinetic energy T, it's expectation value can be calculated using gaussian integrals,

$$\langle T\rangle = \int_{-\infty}^{\infty} \psi^*\Big(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\Big)\psi\, dx = -\frac{\hbar^2}{2m}\int_{-\infty}^{\infty} \psi^*\psi\big[\alpha^2x^2 - \alpha\big]\, dx = -\frac{\hbar^2}{2m}\Big[\frac{\alpha}{2} - \alpha\Big] = \frac{\hbar^2\alpha}{4m}$$

In other words, the expected kinetic energy scales linearly with α.

Calculating the expected potential energy $\langle V \rangle$ is expressed by

$$\langle V \rangle = \int_{-\infty}^{\infty} \psi^* |V(x)| \psi \, dx = \left(\frac{\alpha}{\pi}\right)^{1/2} \int_{-\infty}^{\infty} -V(x) e^{-\alpha x^2} \, dx$$

If we take a suitably small α, $\exp(-\alpha x^2) \approx 0$. Then the expected total energy can be expressed as,

$$\langle H \rangle \approx \left(\frac{\alpha}{\pi}\right)^{1/2} \left[\frac{\hbar^2 (\alpha\pi)^{1/2}}{4m} + \int_{-\infty}^{\infty} V(x) dx \right]$$

Now we can see that if we let α tend towards 0^+, within the parenthesis, the expected kinetic energy will arbitrarily minimize on the order of $\sqrt{\alpha}$ but $V(x)$ is constant and negative. Thus a suitably small α will result in a negligible (positive) kinetic energy term and a dominant (negative) potential energy term.

Further more, given the findings of part a, if $E(\alpha)$ can be made suitably negative, and $0 > E(\alpha) > E_0$ for all α, then there must exist a ground state E_0 that is negative for the attractive potential.

Exercise 5.2.3 Consider $V(x) = -aV_0\delta(x)$. Show that it admits a bound state of energy $E = -ma^2V_0^2/2\hbar^2$. Are there any other bound states?

We begin by solving the time independent schrödinger equation in the position basis.

$$\begin{aligned}
\langle x | \, P^2/2m + -aV_0\delta(x) \, |E\rangle &= \langle x | \, E \, |E\rangle \\
-\frac{\hbar^2}{2m} \langle x | \, D^2 \, |E\rangle - aV_0 \langle x | \, \delta(x) \, |E\rangle &= \langle x | \, E \, |E\rangle \\
-\frac{\hbar^2}{2m} \psi_E''(x) - aV_0\delta(x)\psi_E(x) &= E\psi_E(x) \\
\psi''(x) &= -\frac{2m}{\hbar^2}\Big[E + aV_0\delta(x)\Big]\psi_E(x)
\end{aligned}$$

A boundary condition in the second derivative can be found by integrating the TISE over an infinitessimaly small region ϵ centered around 0.

$$\int_\epsilon \psi_E''(x)dx = -2mE/\hbar^2 \int_\epsilon \psi_E(x)dx - 2maV_0/\hbar^2 \int_\epsilon \delta(x)\psi_E(x)dx$$

The first integral evaluates to zero as epsilon approaches 0, since $\psi_E(x)$ is continuous. Then remaining expression can be thought as the change of the slope of $\psi_E(x)$ across $x = 0$.

$$\psi_E'(\epsilon) - \psi_E'(-\epsilon) = -\frac{2maV_0\psi(0)}{\hbar^2}$$

Following this, our strategy is to solve for $\psi(x)$ at $x \neq 0$. Then, we can use the boundary conditions to determine the expression over the full domaine. In this regard, the case where $E > 0$, the solution is trivially the free particle. For the case where $E < 0$, then,

$$\psi(x) = A\exp(-x\sqrt{2m|E|}/\hbar) + B\exp(x\sqrt{2m|E|}/\hbar)$$

B must be zero to have correct behaviour as $x \to \infty$. I must restrict the function piece wise to do so.

$$\begin{cases} \psi(x) = A\exp(-x\sqrt{2m|E|}/\hbar) & , \ x > 0 \\ \psi(x) = A\exp(x\sqrt{2m|E|}/\hbar) & , \ x < 0 \end{cases}$$

Then A can be set via normalization

$$1 = \int_{-\infty}^{\infty} |\psi|^2 dx = 2\int_{-\infty}^{0} |A|^2 \exp(-2x\sqrt{2m|E|}/\hbar)dx$$

. Thus $A = (2m|E|)^{1/4}/\sqrt{\hbar}$

The $\psi(x)$ we solved for can estimate $\psi'(x=0)$, approaching from both sides. By evaluating the derivative, see that $\psi'(x \to 0^{\pm}) = \mp A\sqrt{2m|E|}/\hbar$. We can use this to express the change in the slope across $x = 0$ and directly equate it to the boundary term from earlier. Solving for —E— we arrive at the desired bound state energy equation.

$$\Delta\psi'(0) = 2A\sqrt{2m|E|}/\hbar = -\frac{2maV_0A}{\hbar^2}$$
$$E = -ma^2V_0^2/2\hbar^2$$

This is the only solution that solves the TISE, thus it is the only bound energy state.

Exercise 5.2.4 Consider a particle of mass m in the state $|n\rangle$ of a box of length L. Find the force $F = -\partial E/\partial L$ encountered when the walls are slowly pushed in, assuming the particle remains in the nth state of the box as its size changes. Consider a classical particle of energy E_n in this box. Find its velocity, the frequency of collision on a given wall, the momentum transfer per collision, and hence the average force. Compare it to $-\partial E/\partial L$ computed above.

To recap, a particle in a box has wave function and energies as follows

$$\begin{cases} \psi_n = \left(\frac{2}{L}\right)^{1/2} \cos\left(\frac{n\pi x}{L}\right), & n \text{ odd} \\ \psi_n = \left(\frac{2}{L}\right)^{1/2} \sin\left(\frac{n\pi x}{L}\right), & n \text{ even} \\ E_n = \hbar^2\pi^2n^2/2mL^2 \end{cases}$$

Thus, for the quantum case, $F_n = 2\hbar^2\pi^2n^2/2mL^3 = 2E_n/L$.

For the classical case, E_n gives it's velocity as $v = \sqrt{2E_n/m}$. Assuming a static wall and a velocity only along $\hat{x}$, the frequency of collision on a given wall is going to be $f = v/2L$. Then a perfectly elastic collision will result in a change in momentum, p, equal to $2p$.

The average force is related to the change in momentum as $\overline{F} = \Delta p/\Delta t = \Delta p f = pv/L = 2E/L$. This is equal to the quantum case.

Exercise 5.2.5 If the box extends from $x = 0$ to L (instead of $-L/2$ to $L/2$), show that $\psi_n(x) = (2/L)^{1/2}\sin(n\pi x/L)$ for $n = Z^+$ and $E_n = \hbar^2n^2\pi^2/2mL^2$.

The solution for $\psi(x)$ from Exercise 5.2.4 is for a potential box between $-L/2$ and $L/2$. The new box is just a translation of those solutions by $+L/2$.

Then, note that $\cos\left(\frac{n\pi(x-L/2)}{L}\right) = \cos\left(n\pi x/L - n\pi/2\right)$ and $\sin\left(\frac{n\pi(x-L/2)}{L}\right) = \sin\left(n\pi x/L - n\pi/2\right)$

Firstly, the energy of the system is invariant under a this translation, which reflects how the expression for energy is equal in both scenarios.

For all even n, the expression $\sin\left(n\pi x/L - n\pi/2\right)$ corresponds to a π or 2π rotation about the original angle $n\pi x/L$. Thus it is equivalent to $\pm|\sin(n\pi x/L)|$. Since wavefunctions that differ by only a global phase are functionally equivalent, then it simplifies to the desired sin expression.

For odd numbers, it corresponds to a rotation $\pi/2$ or $-\pi/2$ about the original angle $n\pi x/L$. Using the phase relation of cosine and sine, this expression is then equivalent to the desired sin expression up to a global phase.

Completing all cases, we arrive at the final expression of $\psi_n(x)$

Exercise 5.2.6 The square well potential. Consider a particle in a square well potential:

$$V(x) = \begin{cases} 0 & ,|x| \leq a \\ V_0 & ,|x| \geq a \end{cases}$$

Since when $V_0 \to \infty$, we have a box, let us guess what the lowering of the walls does to the states. First of all, all the bound states (which alone we are interested in), will have $E \leq V_0$. Second, the wave functions of the low-lying levels will look like those of the particle in a box, with the obvious difference that ψ will not vanish at the walls but instead spill out with an exponential tail. The eigenfunctions will still be even, odd, even, etc.

1. Show that the even solutions have energies that satisfy the transcendental equation $k\tan ka = \kappa$. And the odd ones will have energies that satisfy $k\cot ka = -\kappa$. where k and $i\kappa$ are real and complex wavenumbers inside and outside the well, respectively. Note that k and κ are related by $k^2 + \kappa^2 = 2mV_0/\hbar^2$. Verify that as V_0 tends to ∞ , we regain the levels in the box.

Our strategy is to solve this like how we would for a regular particle in the box. Lets denote the wavefunction as $x \leq |a|$ as ψ_I, then for $x \geq |a|$ it is ψ_{III}. Then we have ψ_{II} inside the box.

We first solve the TISE in the x basis,

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V(x))\psi = 0$$

$$\frac{d^2\psi}{dx^2} - \frac{2m}{\hbar^2}|E - V_0|\psi = 0$$

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}E\psi = 0$$

The equation inside the box is trivially solved by the free particle expression, for $k = \sqrt{2mE}/\hbar$

$$\psi_{II} = Ae^{ikx} + Be^{-ikx}$$

Then outside the box, the expression can be solved, for $\kappa = \sqrt{2m(V_0 - E)}/\hbar$

$$\psi_I = A'e^{\kappa x}$$
$$\psi_{III} = B'e^{-\kappa x}$$

Note that other coefficients to the general solution to ψ_I and ψ_{III} were excluded on physical grounds.

We now enforce that $\psi(|a|)$ with the solutions from inside the box equals that of the solution from outside the box. Additionally, we know that since V contains finite jumps, ψ' is continuous. So we get two more equations evaluated at a.

$$\begin{aligned}
\psi(-a) = Ae^{-ika} + Be^{ika} &= A'e^{-\kappa a} \\
\psi(a) = Ae^{ika} + Be^{-ika} &= B'e^{-\kappa a} \\
\psi'(-a) = -Aike^{-ika} + Bike^{ika} &= -\kappa A'e^{-\kappa a} \\
\psi'(a) = Aike^{ika} - Bike^{-ika} &= -\kappa B'e^{-\kappa a}
\end{aligned}$$

The **even** solutions of these equations satisfy $\psi(x) = \psi(-x)$ and the same for the first derivative.

In this case, $A' = B'$, so

$$\begin{aligned}
(A - B)e^{ika} + (B - A)e^{-ika} &= 0 \\
ik(A - B)e^{ika} - ik(B - A)e^{-ika} &= 0
\end{aligned}$$

Clearly, the boundary conditions are satisfied when $A' = B'$ and $A = B$.

$$\begin{aligned}
\psi(-a) = Ae^{-ika} + Ae^{ika} &= A'e^{-\kappa a} \\
\psi(a) = Ae^{ika} + Ae^{-ika} &= A'e^{-\kappa a} \\
\psi'(-a) = -Aike^{-ika} + Aike^{ika} &= -\kappa A'e^{-\kappa a} \\
\psi'(a) = Aike^{ika} - Aike^{-ika} &= -\kappa A'e^{-\kappa a}
\end{aligned}$$

Or equivalently

$$\begin{aligned}
2A\cos ka &= A'e^{-\kappa a} \\
2Ak\sin ka &= A'\kappa e^{-\kappa a}
\end{aligned}$$

Moving around terms and equating, we arrive at the final desired expression $k\tan ka = \kappa$

The **odd** solutions satsify $\psi(-x) = -\psi(x)$. We then start again from our boundary equations

$$\begin{aligned} Ae^{-ika} + Be^{ika} &= -Ae^{ika} - Be^{-ika} \\ A'e^{-\kappa a} &= -B'e^{-\kappa a} \\ -Aike^{-ika} + Bike^{ika} &= -Aike^{ika} + Bike^{-ika} \\ -\kappa A'e^{-\kappa a} &= \kappa B'e^{-\kappa a} \end{aligned}$$

Thus here, $A' = -B'$ and $A = -B$. Via similar steps to the even case we can see that

$$\begin{aligned} -i2A\sin ka &= A'e^{-\kappa a} \\ ki2A\cos ka &= \kappa A'e^{-\kappa a} \end{aligned}$$

Then by solving for $-\kappa$ we arrive at the desired expression $-\kappa = k\cot ka$

2. The previous even and odd solutions must be solved graphically. In the ($\alpha = ka$, $\beta = \kappa a$) plane, imagine a circle that obeys the expression $k^2 + \kappa^2 = 2mV_0/\hbar^2$. The bound states are then given by the interaction of the curve $\alpha\tan\alpha = \beta$ or $\alpha\cot\alpha = -\beta$ with the circle. (Remember α and β are positive.

Just plug in the equations to a graphing calculator to solve.

3. Show that there is always one even solution and that there is no odd solution unless $V_0 \geq \hbar^2\pi^2/8ma^2$. What is E when V_0 just meets this requirement? Note that the general result from Exercise 5.2.2b holds.

Since for even solutions a wavenumber of $k = 0$ implies $\kappa = 0$, in the graphic view. One can consider how the even solutions pass through the origin, then a circle centered on the origin will always intersect through the trancendental equations, always yielding an even solution. But there exists no intersection unless the radius of the circle is large enough to intersect the first odd transcendental equation which yields the requirement for the odd solution. The answer is found numerically.

Exercise 5.3.1 Consider the case where $V = V_r - iV_i$, where the imaginary part V_i is constant. Is the Hamiltonian Hermitian? Go through the derivation of the continuity equation and show that the total probability for finding the particle decreases exponentially as $e^{-2V_it/\hbar}$. Such complex potentials are used to describe processes in which particles are absorbed by a sink.

Immediately, we see that the Hamiltonina is not hermitian since $V^\dagger = V_r + iV_i \neq V$. We start in the same method carried by the book for the case of the continuity equation for hermitian H, with the Schrödinger equation in 3D.

$$\begin{aligned} i\hbar\dot{\psi} &= -\frac{\hbar^2}{2m}\nabla^2\psi + V\psi \\ -i\hbar\dot{\psi} &= -\frac{\hbar^2}{2m}\nabla^2\psi^* + V^*\psi^* \end{aligned}$$

By multiplying the first equation by ψ^* on the left and the second equation by ψ on the left, then subtracting, we arrive at the following,

$$i\hbar\frac{\partial}{\partial t}\Big(\psi^*\psi\Big) = -\frac{\hbar^2}{2m}\Big[\psi^*\nabla^2\psi - \psi\nabla^2\psi^*\Big] - \psi^*\psi\Big[V - V^*\Big]$$
$$\frac{\partial}{\partial t}\Big(\psi^*\psi\Big) = -\frac{\hbar}{2mi}\nabla\cdot\Big[\psi^*\nabla\psi - \psi\nabla\psi^*\Big] - \psi^*\psi\Big[V - V^*\Big]$$
$$\dot{P} = -\nabla\cdot j - \frac{2V_i}{\hbar}P$$

If we integrate overall space to arrive at the total probability, the integral over $\nabla\cdot j$ drops out, since for typical normalizable wave functions the surface integral of j on an infinitely far surface vanishes (using divergence theorem). After the spatial integration we are left with a differential equation in time $\dot{P_{tot}} = -2V_iP_{tot}/\hbar$, which is solved by the desired decreasing exponential $e^{-2V_it\hbar}$.

Exercise 5.3.2 Convince yourself that if $\psi = c\tilde{\psi}$, where c is a constant (real or complex) and $\tilde{\psi}$ is real, the corresponding j vanishes.

This is simply shown with the definition of j.

$$j = \frac{\hbar}{2mi}(\psi^*\nabla\psi - \psi\nabla\psi^*) = \frac{\hbar}{2mi}|c|(\tilde{\psi}\nabla\tilde{\psi} - \tilde{\psi}\nabla\tilde{\psi}) = 0$$

Exercise 5.3.3 Consider

$$\psi_p = \left(\frac{1}{2\pi\hbar}\right)^{3/2} e^{i(\vec{p}\cdot\vec{r})/\hbar}$$

Find $\vec{j}$ and P and compare the relation between them to the electromagnetic equation $\vec{j} = \rho\vec{v}$, $\vec{v}$ being velocity. Since ρ and $\vec{j}$ are constant, note that the continuity equation is trivially satisfied.

For short hand, let $A = (1/2\pi\hbar)^{3/2}$, and $e^{\pm} = e^{\pm i\vec{p}\cdot\vec{r}/\hbar}$. Then we write down $\vec{j}$

$$\begin{aligned}\vec{j} &= \frac{\hbar}{2mi}\Big(A^2e^-\nabla e^+ - A^2e^+\nabla e^-\Big)\\ &= \frac{\hbar A^2}{2mi}\Big(e^-\Big[(ip_x/\hbar)e^+\vec{i} + (ip_y/\hbar)e^+\vec{j} + (ip_z/\hbar)e^+\vec{k}\Big]\\ &- e^+\Big[(-ip_x/\hbar)e^+\vec{i} + (-ip_y/\hbar)e^+\vec{j} + (-ip_z/\hbar)e^+\vec{k}\Big]\\ &= \frac{A^2}{m}\vec{p}\end{aligned}$$

Exercise 5.3.4 Consider $\psi = Ae^{ipx/\hbar} + Be^{-ipx/\hbar}$ in one dimension. Show that $j = (|A|^2 - |B|^2)p/m$. The absence of cross terms between the right- and left-moving pieces in ψ allow us to associate the two parts of j with corresponding parts of ψ.

Once again we start with the formula for $\vec{j}$ and use the short hand $e^{\pm} = e^{\pm ipx/\hbar}$

$$\begin{aligned}\vec{j} &= \frac{\hbar}{2mi}(\psi^*\nabla\psi - \psi\nabla\psi^*)\\ \vec{j} &= \frac{\hbar}{2mi}((A^*e^{-ipx/\hbar} + B^*e^{ipx/\hbar})\nabla(Ae^{ipx/\hbar} + Be^{-ipx/\hbar}))\\ &- (Ae^{ipx/\hbar} + Be^{-ipx/\hbar}))\nabla(A^*e^{-ipx/\hbar} + B^*e^{ipx/\hbar}))\end{aligned}$$

After taking the gradients, a common product of $\pm ip/\hbar$is pulled out of every term (from the gradient acting on $e^{\pm}$. Note that the cross terms with coefficients A^*B and AB^* cancel. Then non-cross terms have phases that cancel each other out to only leave the modulus of the coefficients A and B. The double counting of these coefficients is also canceled with the $1/2$ coefficient the entire coefficient has. What remains is the desired expression.

$$\vec{j} = (|A|^2 - |B|^2)p/m$$

Exercise 5.3.4 Evaluate the third piece in Eq. (5.4.16) and compare the resulting T with Equation. (5.4.21). Hint: Expand the factor $(k_1^2 - 2mV_0/\hbar^2)^{1/2}$ near $k_1 = k_0$, keeping just the first derivative in the Taylor series.